

Protiva Rahman

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EDUCATION

The Ohio State University, Columbus, Ohio

Ph.D. in Computer Science and Engineering, 2020

B.S. in Computer and Information Science, Minor in Mathematics, *cum laude*, 2015

ACADEMIC APPOINTMENTS

University of Florida College of Medicine, Health Outcomes and Biomedical Informatics

Research Assistant Professor (*June 2023–present*)

Vanderbilt University Medical Center, Biomedical Informatics

Adjunct Assistant Professor (*June 2023–present*)

RESEARCH AND TEACHING EXPERIENCE

Vanderbilt University Medical Center, Biomedical Informatics

Postdoctoral Research Fellow (*September 2020–May 2023*), **PI**: Dr. Daniel Fabbri

- Extracting Unstructured Data from Electronic Health Records and Biomedical Literature

National Library of Medicine, National Center for Biotechnology Information

Scientific Visitor (*May 2019 - August 2019*), **PI**: Dr. Zhiyong Lu

- User Evaluation of Biomedical Literature Retrieval

The Ohio State University, Computer Science and Engineering

Graduate Research Associate (August 2015 – August 2020), **PI**: Drs. Arnab Nandi and Courtney Hebert

- Amplifying Domain Expertise in Medical Data Pipelines

The Ohio State University, Computer Science and Engineering

Grader for Introduction to Programming in Java (*January 2013–April 2013*)

- Held office hours and graded homework assignments and lab projects for 45 students

PUBLICATIONS

1. **Protiva Rahman**, Michele LeNoue-Newton, Kenneth Kehl et al. Clinical and Genomic Predictors of Central Nervous System Metastases in Non-Small Cell Lung Cancer Patients: An AACR Project GENIE Study. *Under Review*.
2. Jeremy Balch, Sasank Desaraju, Victoria Nolan, Divya Vellanki, Timothy Buchanan, Lindsey Brinkley, Yordan Penev, Ahmet Bilgili, Aashay Patel, Corinne Chatham, David Vanderbilt, Rayon Uddin, Azra Bihorac, Philip Efron, Tyler Loftus, **Protiva Rahman**, Benjamin Shickel. Language Models for Multilabel Document Classification of Surgical Concepts in Exploratory Laparotomy Operative Notes: Algorithm Development Study. *JMIR Medical Informatics*. 2025 Jul 9;13:e71176.
3. Levente Lippenszky, Kathleen F Mittendorf, Zoltán Kiss, Michele L LeNoue-Newton, Pablo Napan-Molina, **Protiva Rahman**, Cheng Ye, Balázs Laczi, Eszter Csernai, Neha M Jain, Marilyn E Holt, Christina N Maxwell, Madeleine Ball, Yufang Ma, Margaret B Mitchell, Douglas B Johnson, David S Smith, Ben H Park, Christine M Micheel, Daniel Fabbri, Jan Wolber, Travis J Osterman. Prediction of Effectiveness and Toxicities of Immune Checkpoint Inhibitors Using Real-World Patient Data. *JCO Clinical Cancer Informatics* 2024.
4. Megan M Shuey, Yihua Wang, Rachel R Xiang, Aaron Zou, **Protiva Rahman**, Daniel Fabbri, Joshua A Beckman, Iris Z Jaffe, Quinn S Wells. Aggregation and Contextualization of Murine Investigations Improves Discovery of Significant Human Atherosclerotic Cardiovascular Disease Associations. *Circulation* 2024.

5. **Protiva Rahman**, Cheng Ye, Kathleen Mittendorf, Michele LeNoue-Newton, Christine Micheel, Jan Wolber, Travis Osterman, Daniel Fabbri. Accelerated Curation of Checkpoint Inhibitor-Induced Colitis Cases from Electronic Health Records. *Journal of American Medical Informatics Association Open* 2023.
6. **Protiva Rahman**, Daniel Fabbri. Semi-Automated Data Curation from Biomedical Literature. *Proceedings of American Medical Informatics Association* 2022.
7. Megan M Shuey, Rachel R Xiang, M Elizabeth Moss, Brigett V Carvajal, Yihua Wang, Nicholas Camarda, Daniel Fabbri, **Protiva Rahman**, Jacob Ramsey, Alec Stepanian, Paola Sebastiani, Quinn S Wells, Joshua A Beckman, Iris Z Jaffe. Systems Approach to Integrating Preclinical Apolipoprotein E-Knockout Investigations Reveals Novel Etiologic Pathways and Master Atherosclerosis Network in Humans. *Arteriosclerosis, thrombosis, and vascular biology* 2022.
8. **Protiva Rahman**, Arnab Nandi, Courtney Hebert. Amplifying Domain Expertise in Medical Data Pipelines. *Journal of Medical Internet Research Medical Informatics* 2020.
9. **Protiva Rahman**, Lilong Jiang, Arnab Nandi. Evaluating Interactive Data Systems: Survey and Case Studies. *The VLDB Journal* 2020.
10. Courtney Hebert, Yuan Gao, **Protiva Rahman**, Courtney Dewart, Mark Lustberg, Preeti Pancholi, Kurt Stevenson, Nirav Shah, Erinn M Hade. Prediction of antibiotic susceptibility for urinary tract infection in a hospital setting. *Antimicrobial Agents and Chemotherapy* 2020.
11. **Protiva Rahman**, Arnab Nandi. TRANSFORMER: A Database-Driven Approach to Generating Forms for Constrained Interaction. *Proc. of the 24th International Conference on Intelligent User Interfaces* 2019 (acceptance rate: 25%).
12. **Protiva Rahman**, Courtney Hebert, Arnab Nandi. ICARUS: Minimizing Human Effort in Iterative Data Completion. *Proc. of the VLDB Endowment* 2018 (acceptance rate: 17%).
13. **Protiva Rahman**, Jian Chen, Courtney Hebert, et al. Exploratory Visualizations of Rules for Validation of Expert Decisions. *3rd Workshop on Data Systems for Interactive Analysis, Proceedings of the IEEE Visualization Conference* 2018.
14. Courtney Dewart, Yuan Gao, **Protiva Rahman**, et al. Penicillin Allergy and Association with Ciprofloxacin Coverage in Community-onset Urinary Tract Infection. *Journal of Infection Control & Hospital Epidemiology* 2018.

CONFERENCE PRESENTATIONS

1. **Protiva Rahman**, Cheng Ye, Kate Mittendorf, Michele LeNoue-Newton, Christine Micheel, Daniel Fabbri. Accelerated Data Curation of Colitis Cases. *AMIA* 2022.
2. **Protiva Rahman**, Daniel Fabbri. Automatic Data Curation from Unstructured Text. *AMIA* 2021.
3. **Protiva Rahman**, Michele LeNoue-Newton, Sandip Chaugai, et al. Clinical and genomic predictors of brain metastases (BM) in non-small cell lung cancer (NSCLC): An AACR Project GENIE analysis. *ASCO* 2021.
4. **Protiva Rahman**, Erinn Hade, Emily Patterson, Courtney Hebert. Amplifying Domain Expertise in Antimicrobial Data Pipeline. *AMIA Oral Presentation* 2020.
5. **Protiva Rahman**, Elisabeth Root, Susan D. Moffatt-Bruce, Courtney Hebert. Designing an Interface for a Hospital Geographic Information System. *AMIA* 2020.
6. **Protiva Rahman**, Erinn Hade, Courtney Dewart et al. Predicting Empiric Antibiotic Coverage based on Patient Factors. *AMIA* 2019.
7. **Protiva Rahman**, Courtney Hebert, Arnab Nandi. Enabling Effective Data Interaction for Domain Experts. *IEEE ICHI PhD Symposium* 2018.
8. **Protiva Rahman**, Erinn Hade, Arnab Nandi et al. Derivation of Expert Consensus Rules for Missing Antimicrobial Susceptibility Data. *AMIA* 2018.
9. Lilong Jiang, **Protiva Rahman**, Arnab Nandi. Evaluating Interactive Data Systems: Workloads, Metrics, and Guidelines. *SIGMOD 2018 Tutorial*.

10. **Protiva Rahman**, Courtney Hebert, Albert Lai. Parsing Complex Microbiology Data for Secondary Use. *AMIA 2016*.
11. Kathleen F Mittendorf, Christine Micheel, Michele LeNoue-Newton, **Protiva Rahman**, Daniel Fabbri, Jan Wolber, Travis John Osterman, Mohamed Fouda, Ben Ho Park. Overcoming barriers in academic-industry partnerships to improve predictive modeling in immuno-oncology. *ASCO 2021*.

PATENT APPLICATIONS

- Rewriting Forms for Constrained Interaction. USPTO: 15/855,515

INVITED TALKS

- Vanderbilt University, “Superheroes for Superbugs”, Feb 2020.
- UC Berkeley, “Amplifying Domain Expertise to Combat Antimicrobial Resistance”, Jan 2020.
- University of Southern California. “Amplifying Domain Expertise in Data Pipelines”, May 2019.

INDUSTRY EXPERIENCE

Nationwide Financial, NFIT Reporting and Data Services

Application Developer Intern (*May 2014-August 2014*)

- Developed SQL reports and Informatica mappings for NF sales

Dahbashi Engineering UAE

Engineering Intern (*August 2010 - September 2010*)

- Improved company’s Oracle management information system

AWARDS

- SIGMOD Student Travel Award (*2018*)
- IEEE ICHI Doctoral Consortium Student Travel Award (*2018*)
- OSU WEGC Outstanding Council Member (*2018*)
- Marianna Russell Technology Grant (*2017*)
- CRA-W Grad Cohort (*2017*)
- Data Quality in the Era of Big Data Workshop Travel Fellowship (*2016*)
- ACM-W Grace Hopper Travel Scholarship (*2016*)

LEADERSHIP AND SERVICE

- PC Member: EDML 2019, HILDA 2023, SIGMOD 2025-2026
- Reviewer: JAMIA 2023, ACM CHI 2020, AMIA 2019,2021-2023, ACM TODS 2019, VLDBJ 2025
- AMIA Conference First Look Student Panel 2020
- IUI Conference Student Volunteer 2019
- Women in Engineering Graduate Council, *Member:2017, SWE Rep:2018, VP:2019*
- A Kid Again, *Awareness Chair (2012 – 2014)*
- International Student Organization, *Secretary (2012 – 2013)*
- Phi Beta Kappa
- National Residence Hall Honorary

ADDITIONAL WORK EXPERIENCE

The Ohio State University Housing, Barrett Complex

Resident Manager (*August 2014-April 2015*)

- Managed a 24-hour desk which served the residents and guests of a 350-person residence
- Hired, trained, and supervised a staff of 17 student employees
- Conducted staff meetings with a focus on professional development and diversity and inclusion

The Ohio State University Housing, Morrill Tower

Resident Advisor (*September 2012-April 2014*)

- Held proactive conversations with residents to assist with their academic needs and concerns
- Organized programs to build a safe, fun and close-knit community for 69 residents

Le Meridien Dubai, UAE

Sales and Marketing Intern (*May 2013 – June 2013*)

- Interacted with magazine editors to build media relations
- Persuaded clients to increase business with the hotel during sales calls

SKILLS

- Python, R, Java, C++, C, SQL, HTML, Javascript, CSS, Object Oriented Programming
- Fluent in English, Bengali, Hindi and intermediate knowledge of French